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Telemedical Diabetic Retinopathy Screening in a Primary Care Setting: Quality of Retinal Photographs and Accuracy of Automated Image Analysis

Maximilian W.M. Wintergerst^{ID*}, Veronica Bejan^{a*}, Vera Hartmann^a, Marina Schnorrenberg^b, Markus Bleckwenn^c, Klaus Weckbecker^b, and Robert P. Finger^{ID^a}

^aDepartment of Ophthalmology, University Hospital Bonn, Bonn, Germany; ^bInstitute of General Practice and Interprofessional Care, Faculty of Health/Department of Medicine, University Witten/Herdecke, Witten, Germany; ^cDepartment of General Practice, Medical Faculty, University of Leipzig, Leipzig, Germany

ABSTRACT

Background: Screening for diabetic eye disease (DED) and general diabetes care is often separate, which leads to delays and low adherence to DED screening recommendations. Thus, we assessed the feasibility, achieved image quality, and possible barriers of telemedical DED screening in a point-of-care general practice setting and the accuracy of an automated algorithm for detection of DED.

Methods: Patients with diabetes were recruited at general practices. Retinal images were acquired using a non-mydriatic camera (CenterVue, Italy) by medical assistants. Images were quality assessed and double graded by two graders. All images were also graded automatically using a commercially available artificial intelligence (AI) algorithm (EyeArt version 2.1.0, Eyenuk Inc.).

Results: A total of 75 patients (147 eyes; mean age 69 years, 96% type 2 diabetes) were included. Most of the patients (51; 68%) preferred DED screening at the general practice, but only twenty-four (32%) were willing to pay for this service. Images of 63 patients (84%) were determined to be evaluable, and DED was diagnosed in 6 patients (8.0%). The algorithm's positive/negative predictive values (95% confidence interval) were 0.80 (0.28–0.99)/1.00 (0.92–1.00) and 0.75 (0.19–0.99)/0.98 (0.88–1.00) for detection of any DED and referral-warranted DED, respectively.

Overall, the number of referrals was 18 (24%) for manual telemedical assessment and 31 (41%) for the artificial intelligence (AI) algorithm, resulting in a relative increase of referrals by 72% when using AI.

Conclusions: Our study shows that achieved overall image quality in a telemedical GP-based DED screening was sufficient and that it would be accepted by medical assistants and patients in most cases. However, good image quality and integration into existing workflow remain challenging. Based on these findings, a larger-scale implementation study is warranted.

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Introduction

Worldwide, 425 million adults have type 2 diabetes mellitus (DM).¹ Due to rapid transitions of lifestyles conducive to type 2 DM development, prevalence increases considerably globally, outpacing concurrent development of necessary healthcare services in many countries.² With increasing type 2 DM prevalence, complications such as diabetic eye and/or kidney disease also become more common. Due to this, the World Health Organization (WHO) recently classified diabetic eye disease (DED), comprising diabetic retinopathy and diabetic maculopathy, as a priority eye disease.³ DED is the most common cause of blindness and severe visual loss in working age adults worldwide, and DED affects

almost 80% of those with type 2 DM after 10 years of disease.⁴ Highly effective treatments are available to prevent DED-related visual loss, such as retinal laser photocoagulation and intravitreal injection therapy.⁴ However, effectiveness of these treatments depends on early diagnosis. As early DED stages remain asymptomatic, DED screening in persons with DM is crucial.⁴ Countries with nationwide screening programs have successfully reduced blindness caused by DED by up to 50%.⁵ While screening for DED at population level has not been found to be cost-effective, integration of DED screening into primary or diabetes care has been found to be highly cost-effective.^{6,7}

Thus, it is self-evident that DED screening is necessary as well as a good investment. However, to date,

CONTACT Robert P. Finger  robertfinger@gmx.net  Ophthalmic Epidemiology and Neuroretinal Imaging, Department of Ophthalmology, University of Bonn, Ernst-Abbe-Straße 2, Bonn 53127, Germany

*These authors contributed equally.

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DED screening is often fragmented with multiple health providers such as GPs, endocrinologists, optometrists, and ophthalmologists involved, which translates into multiple visits for patients and many opportunities for delays and miscommunication between health providers and patients. In fact, we found, e.g., an unacceptably long delay of 2 years or longer between initial type 2 DM diagnosis and first DED screening in half of all affected persons in Germany when guidelines stipulate immediate DED screening upon diagnosis.⁸ Similarly, we found that coordination between diabetes and eye clinics can be poor and that patients resent having to come for multiple lengthy appointments which are often associated with considerable personal costs (transport, lost wages, etc.).⁹ Against this background, WHO has advocated for integrated DED screening and 'one point-of-care' solutions.¹⁰ In addition, DED screening can be easily delegated to non-ophthalmologists, which saves resources and frees up much-needed ophthalmologist time to be spent on, e.g., DED treatment. Thus, there is a growing need for innovative and effective strategies for cost-effective and integrated DED screening allowing for early detection and prevention of vision loss.^{11–13}

Several studies have explored the use of non-mydratric retinal cameras and telemedicine approaches to DED screening, which can then be integrated into, e.g., primary care.^{14–20} Obtaining retinal images required for DED screening can be delegated to paramedical staff and graded by either trained non-physicians or ophthalmologists. Overall, studies report this to be feasible and well accepted by involved staff.^{13,21–25} Despite this, there remains a shortage of studies evaluating the feasibility and implications of integrating DED screening into primary care and delegation to staff unfamiliar with the eye and ophthalmic imaging on image quality and thus diagnostic accuracy. Furthermore, evaluation of commercially available automated image analysis algorithms for the detection and quantification of DED using images acquired through DED screening services integrated into primary care is warranted. Thus, in this study, we evaluated both the feasibility of telemedical DED screening integrated into primary care practices followed by automated image analysis using commercially available software.

Materials and methods

The study was conducted at the University Hospital Bonn and affiliated family medicine practices in the community (referred to as GP practices). The study followed the tenets of the Declaration of Helsinki and was approved by the Research Ethics Committee of the University of Bonn (approval number 237/17). All

participants gave their written informed consent prior to study inclusion.

Subject recruitment

Participants were recruited from the participating practices' registries and enrolled at their annual or half-yearly or quarterly Disease Management Programme (DMP) Diabetes check-up visits. This DMP includes a clinical assessment by the GP as well as a screening for diabetic nephropathy and diabetic foot syndrome by a diabetic nurse. Enrollment took place between April and August 2018. None of the participating practices performed DED screening, as it is not part of primary care in Germany. Inclusion criteria were a diagnosis of type 1 or type 2 DM. Exclusion criteria were any physical or mental conditions, which – as assessed by the managing GP – made participation in the study or acquisition of retinal images impossible. Ocular exclusion criteria were opacities of any of the optical media (cornea, lens, and vitreous) that prevented the acquisition of retinal images and any ocular surgery in the last two months. Demographic data, medical history, and history of DM were collected for each patient using a standardized questionnaire. Feedback regarding the image acquisition and screening process of both the patients and medical assistants was assessed using a standardized questionnaire. Patients' satisfaction with the DED screening provided in this study and willingness to pay were also assessed. Feedback collected from medical assistants included difficulty in using the device, usefulness, and opportunities for integration of telemedical DED screening in the practice workflow as well as any general problems anticipated or actual problems during image acquisition.

Image acquisition and manual image analyses

Retinal images were acquired with a 45° non-mydratric retinal camera (The 'Digital Retinography System' [DRS], CenterVue, Padua, Italy) without pupil dilation, which allows for fast image acquisition and is well tolerated by patients.²⁶ After initiation of the examination, the device automatically identifies the patient's eye, centers the camera on the pupil, and acquires images according to a pre-specified imaging protocol. Fourteen medical assistants were trained for an hour by an orthoptist experienced in imaging with the DRS in acquisition of non-mydratric retinal images and the operation of the device including digital image transfer for grading. Image acquisition was performed in a darkened room. If dimming was not possible, the participant's head and the device were covered with

a light-tight fabric to reduce ambient light. Both eyes were imaged using the standard 3-field protocol, with images centered on the macula, optic disc, and temporal of the macula (Supplemental Figure 1). If necessary, image acquisition was repeated in order to achieve the best possible image quality. Following image acquisition, pseudonymized images were transferred digitally to a central database by the medical assistants. Image quality was manually graded for illumination, reflex artifacts, peripheral light artifacts, dust particles, smudge, eyelashes, and defocus (Figure 1 and Supplemental Figure 2). Overall image quality was evaluated on a 6-step ordinal score based on how much of the image was evaluable (nothing evaluable, < 25% of the image evaluable, 25–50% evaluable, 50–75% evaluable, 75–90% evaluable, and > 90% evaluable). Image quality was graded by a trained grader, and DED was graded by both a trained grader and an ophthalmologist experienced in retinal image analysis and DED grading. In case

of discrepancies between the two graders, a second ophthalmologist experienced in retinal image analysis and DED grading was consulted to adjudicate. Graders were masked to the patients' clinical data and the results of the automated algorithm. As non-mydratic single-field fundus photography is sufficient for DR screening,^{27,28} eyes were graded “non-evaluable” by the ophthalmologist, if the central 45° macula area was not gradable. All image analyses were performed on a 27-in. monitor, with a display resolution of 1920 × 1080 pixels. In any case of uncertainty, it was zoomed in the image. Following the grading, both the patient and the general practitioner received a letter regarding DED screening results and, if applicable, any incidental findings.

Classification of diabetic eye disease

DED was classified according to the International Clinical Diabetic Retinopathy and Diabetic Macular

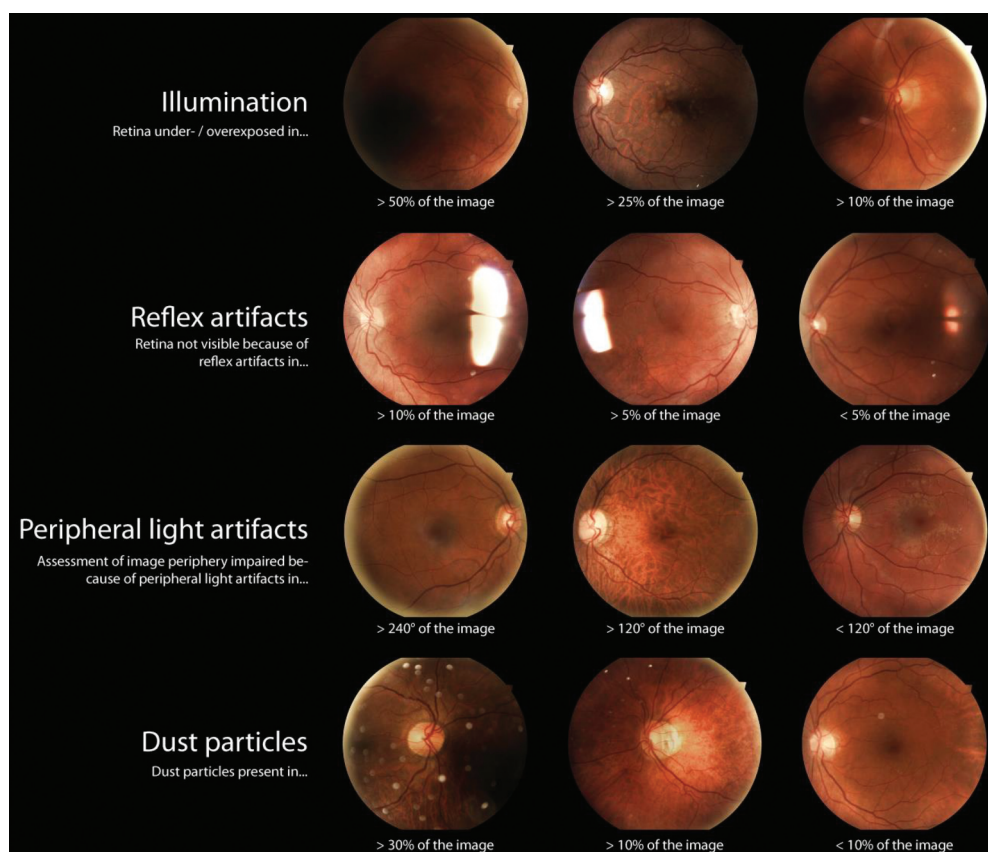


Figure 1. Semi-quantitative scales for grading of illumination, reflex artifacts, peripheral light artifacts and dust particles. Image quality in terms of illumination, reflex artifacts, peripheral light artifacts and dust particles was graded on 4-step scales: **Illumination:** 1 = >50% under/ overexposed, 2 = >25% under/ overexposed, 3 = >10% under/ overexposed, and 4 = <10% under/overexposed; **Reflex artifacts:** 1 = reflex artifacts in > 10% of the image, 2 = reflex artifacts in > 5% of the image, 3 = reflex artifacts in < 5% of the image, and 4 = no reflex artifacts; **Peripheral light artifacts:** 1 = peripheral light artifacts in > 240° of the image margin, 2 = peripheral light artifacts in > 120° of the image margin, 3 = peripheral light artifacts in < 120° of the image margin, and 4 = no peripheral light artifacts; **Dust particles** 1 = dust particles present in > 30% of the image, 2 = dust particles present in > 10% of the image, 3 = dust particles present in < 10% of the image, and 4 = no dust particles present.

Edema Disease Severity Scales.²⁹ Referral-warranted DED was defined as moderate non-proliferative diabetic retinopathy or worse or presence of clinically significant macular edema.^{30,31} Classification was performed on patient level, that is, if one eye was graded non-evaluable, the patient was classified non-evaluable. Images were double-graded.

Automated image analyses

All images were analyzed using commercially available automated image analysis software (EyeArt, version 2.1.0, EyeNuk Inc, Los Angeles, CA, USA). This cloud-based software has CE marking in the European Union as a Class IIa medical device and has been evaluated for retinal images from different devices.^{32,33} The software classifies DED according to the same international scales used for manual grading, for both the classification of overall DED and the classification of referable DED.^{29,33}

Statistical analyses

Data were first analyzed descriptively using mean, standard deviation, median, minimum, and maximum for continuous data and absolute and relative frequencies for categorical data. Associations of the image quality score with age, the examiner, image location (central,

Table 1. Characteristics of the study population.

	Mean $\pm$ SD or N(%)	Median (range)
Age (years)	68.95 $\pm$ 11.41	71 (32–86)
Sex (male)	37 (49.3%)	
Sex (female)	38 (50.7%)	
Diabetes Type 1	3 (4%)	
Diabetes Type 2	72 (96%)	
Systolic blood pressure in mmHg	138.84 $\pm$ 18.01	138 (100–200)
Diastolic blood pressure in mmHg	80.28 $\pm$ 9.57	80 (60–110)
Fasting glucose level in mg/dl	144.83 $\pm$ 34.14	147 (92–233)
Glycosylated hemoglobin level in %	7.09 $\pm$ 0.97	7 (5.1–9.9)

SD = standard deviation; Min = minimum; Max = maximum; N = number.

nasal, and temporal), and eye laterality (right, left) were assessed in a multiple linear regression model. The presence/absence and severity of DED were calculated as absolute and relative frequencies, and the agreement between the two graders (ophthalmologist and trained grader) in the manual grading and with AI was evaluated using weighted kappa. Algorithm sensitivity and specificity were compared for the subset of images, which was determined evaluable by the ophthalmologist, and the subset, which was determined evaluable by the algorithm. All statistical analyses were performed using the software IBM-SPSS for Windows v. 26.0 or R (R: A Language and Environment for Statistical Computing, R Core Team, R Foundation for Statistical Computing, Vienna, Austria, 2016). The level of statistical significance was set at $p < .05$.

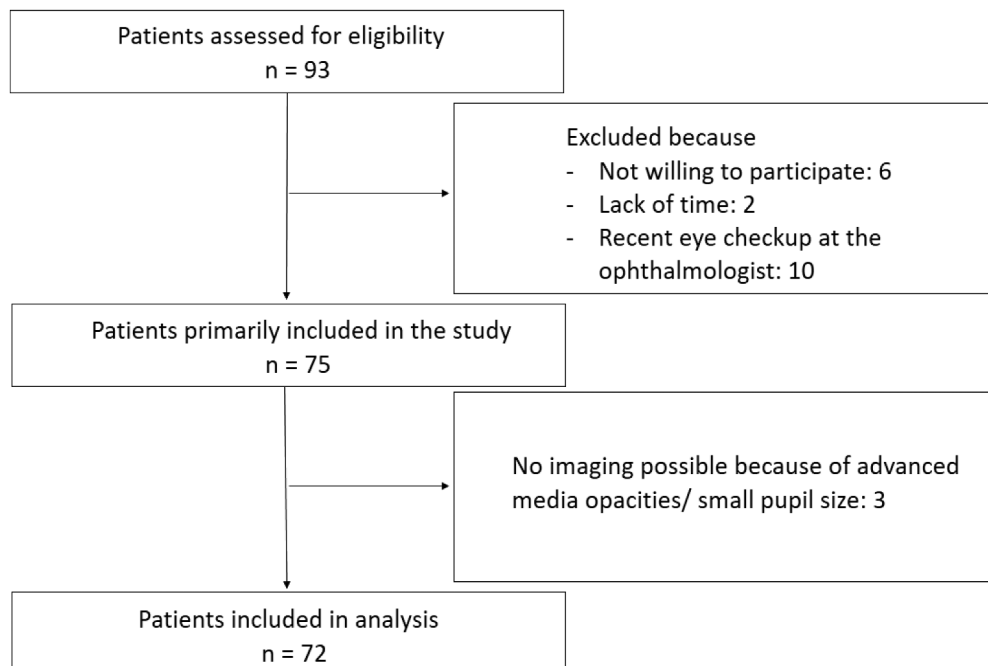


Figure 2. Study flow chart. Inclusion criteria were a diagnosis of type 1 or type 2 DM. Exclusion criteria were any physical or mental conditions, which – as assessed by the managing GP – made participation in the study or acquisition of retinal images impossible. Ocular exclusion criteria were opacities of any of the optical media (cornea, lens, and vitreous) that prevented the acquisition of retinal images and any ocular surgery in the last two months.

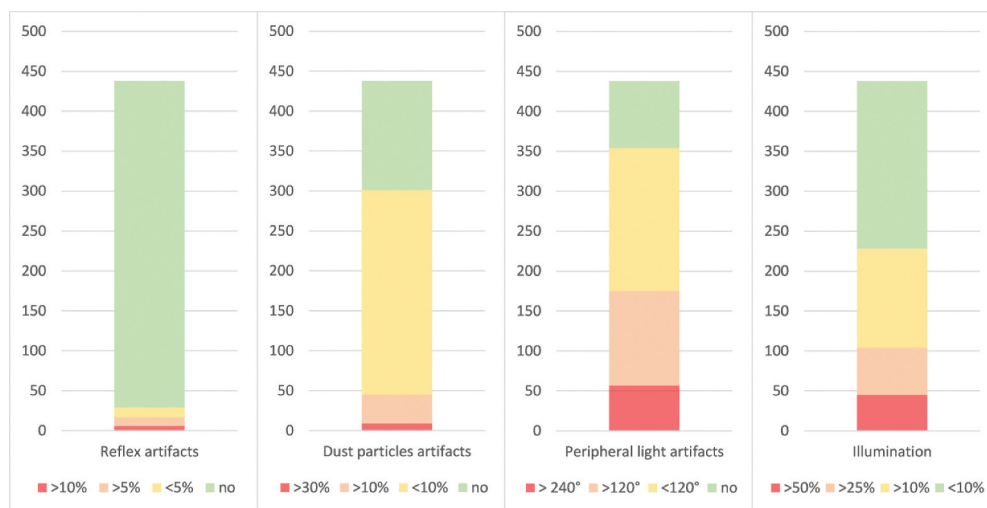


Figure 3. Absolute frequency (the respective amount of images is depicted on the y-axis) of reflex, dust particles and peripheral light artifacts and level of illumination.

Results

Demographics and clinical characteristics

A total of 75 patients (147 eyes) were included in the analysis (Table 1 and Figure 2). Three eyes (corresponding to 6 retinal images) were excluded due to advanced media opacity or too small pupil size. Diabetes duration was up to 5 years in 22%, 6–10 years in 32%, 11–20 years in 31%, and 21 or more years in 11%.

Results of the patients' and examiners' questionnaire

Fifty patients (66.7%) were “very satisfied,” with the DED screening. Most of the patients (51; 68%) would prefer to have their DED screening carried out at the primary care practices, and about one fourth (20; 26.7%) would prefer to have it done at the eye care provider. Twenty-four (32%) patients would be willing to pay on average 28 € (standard deviation $\pm$ 24.87 €; range 5–100 €) for this test. Forty-seven patients refused to answer this question. All medical assistants reported that the retinal imaging device used in this study was easy to use, and just over half (8; 57%) reported that the DED screening could be integrated into their practice's workflow. Reasons given for inability to integrate the DED screening were too much work (3; 21.4%) or too time-consuming (1; 7.1%), with the majority (9; 64.3%), not stating any specific reasons. The majority (10; 71.4%) of the medical assistants found the DED screening useful for their patients.

Image acquisition and quality

A total of 438 retinal images were obtained and graded. In the case of multiple available images for the same imaging location, only the best quality image was retained, leading to 23 images being excluded. An unknown number of images were automatically replaced due to the camera settings, as every new attempt of acquiring an image led to the previous image being overwritten as previously reported.³³ The acquired image location was automatically determined by the non-mydratic retinal camera and 391 of the images (89%) coincided with the actual location determined by manual grading. Image acquisition took on average 7.5 minutes (range 3–16 minutes) for both eyes and the majority (64; 85.3%) complete imaging in less than 10 minutes. The number of attempts to obtain images varied from 1 to 5, with 81.3% of patients successfully completing imaging in three or fewer attempts. Overall image quality was very good, with 83% of images being graded 4 or better. Frequency of reflex, dust particles, and peripheral light artifacts and illumination are illustrated in Figure 3. The majority of images were in focus (91.6%), half of the images exhibited (mostly minor) smudge artifacts (47.7%), and eyelashes artifacts were present in 94 images (21.5%). In multiple linear regression analysis, reduced overall image quality was associated with higher age and temporal image location (Supplemental Table 1 and 2).

Clinical findings and manual grading

Of the 75 examined patients, 63 (84%) were determined evaluable by the ophthalmologist and any DED was

Table 2. Classification of diabetic retinopathy.

	Ophthalmologist N (%)	Algorithm N (%)
No NPDR	57 (76%)	43 (57%)
Mild NPDR	1 (1.3%)	1 (1.3%)
Moderate NPDR	4 (5.3%)	3 (4.0%)
Severe NPDR	1 (1.3%)	1 (1.3%)
PDR	-	-
DME	1 (1.3%)	2 (2.7%)
Non-evaluable	12 (16%)	27 (36%)
Total	75	75

DME = diabetic macular edema; PDR = proliferative diabetic retinopathy;
NPDR = non proliferative diabetic retinopathy.

diagnosed in 6 patients (8.0%). Inter-rater agreement with the trained grader was good with a kappa value of 0.85. Based on patient interviews, DED was already previously documented by a general ophthalmologist and under regular follow-up in 4 patients at the time of the screening. Clinical findings are shown in Table 2.

Comparison with the artificial intelligence algorithm

Twenty-seven patients (36%) were determined “non-evaluable” by the algorithm, whereas only 12 patients (16%) were classified non-evaluable by the ophthalmologist (Table 2), leading to poor agreement (kappa = 0.44, $p < .0001$). In a minority of cases, the image quality grading by the algorithm was not understandable from a human perspective (see Figure 4). Algorithm sensitivity and specificity were compared for the subset of images, which was determined evaluable by the ophthalmologist and the subset, which was determined evaluable by the algorithm (Table 3). The algorithm’s positive/negative predictive values (95% confidence interval) were 0.80 (0.28–0.99)/1.00 (0.92–1.00) and 0.75 (0.19–0.99)/0.98 (0.88–1.00) for detection of any DED and referral-warranted DED, respectively. Overall, the number of referrals due to referable DED or non-evaluable images was 18 (24%) for manual telemedical assessment and 31 (41%) for the artificial intelligence (AI) algorithm, resulting in a relative increase of referrals by 72% when using AI.

Discussion

Our study provides a standardized and systematic approach for image quality grading of non-mydratic retinal images for this camera, which can likely be used and adapted for any other non-mydratic retinal cameras. We showed that achieved overall image quality in a telemedical GP-based DED screening was sufficient and which image quality aspects could still be improved. We could also demonstrate that GP-based telemedical

DED screening was acceptable to medical assistants and patients in the majority of cases. However, there remain relevant constraints. Required training was minimal using a semi-automated image acquisition device. However, challenges concerning mostly image quality as well as integration into existing workflow and services remain. Utilizing an available and CE certified automated image analysis software, referral rates were much higher compared to manual grading, again largely due to suboptimal image quality and non-evaluable images. Based on these findings, a larger-scale implementation study is warranted. Our results emphasize that care should be taken in any cases of automatic image analysis and its interpretation. Our study provides means for standardized image quality assessment, which is an important prerequisite for generation of high-level datasets and comparability of studies.

Novel approaches to DED screening are needed, as the number of persons with diabetes increases while the number of eye healthcare providers and/or their capacity to provide services remains stable or is decreasing, e.g., in Germany.³⁴ One point-of-care solution has been shown to increase acceptance and uptake of healthcare services as well as long-term adherence to both screening and treatment plans.^{35,36} Many persons with diabetes do not require to be seen and/or treated by an ophthalmologist for much of the duration of their diabetes, but require annual or biannual DED screening, which can be delegated to non-ophthalmologists. In fact, this has been done very successfully in a number of countries with centralized healthcare systems such as the United Kingdom.³⁷ Outsourcing DED screening to a national screening program has led, e.g., to a remarkable reduction in sight loss associated with DEDs over the last decade in England.⁵ Countries with no centralized healthcare system will find it inherently more difficult to create a centralized or national screening program, but can utilize e.g., existing disease management programs for diabetes or integrate DED screening services into primary healthcare, which has also been shown to be effective.³⁸ A possible barrier to telemedical DED screening can be patients’ reservations as shown by our study in line with previous findings.^{39,40} Possible reasons include concerns regarding privacy, the potential lack of a relationship with a provider, and reservations regarding accuracy and depth of detail of the telemedical examination reporting.^{39,40}

Artificial intelligence affords us many opportunities including automated retinal image analysis for DED. To what extent this will replace or supplement human graders is difficult to foresee to date and will depend on the technical capabilities of algorithms. In this study, the employed, commercially available algorithm referred

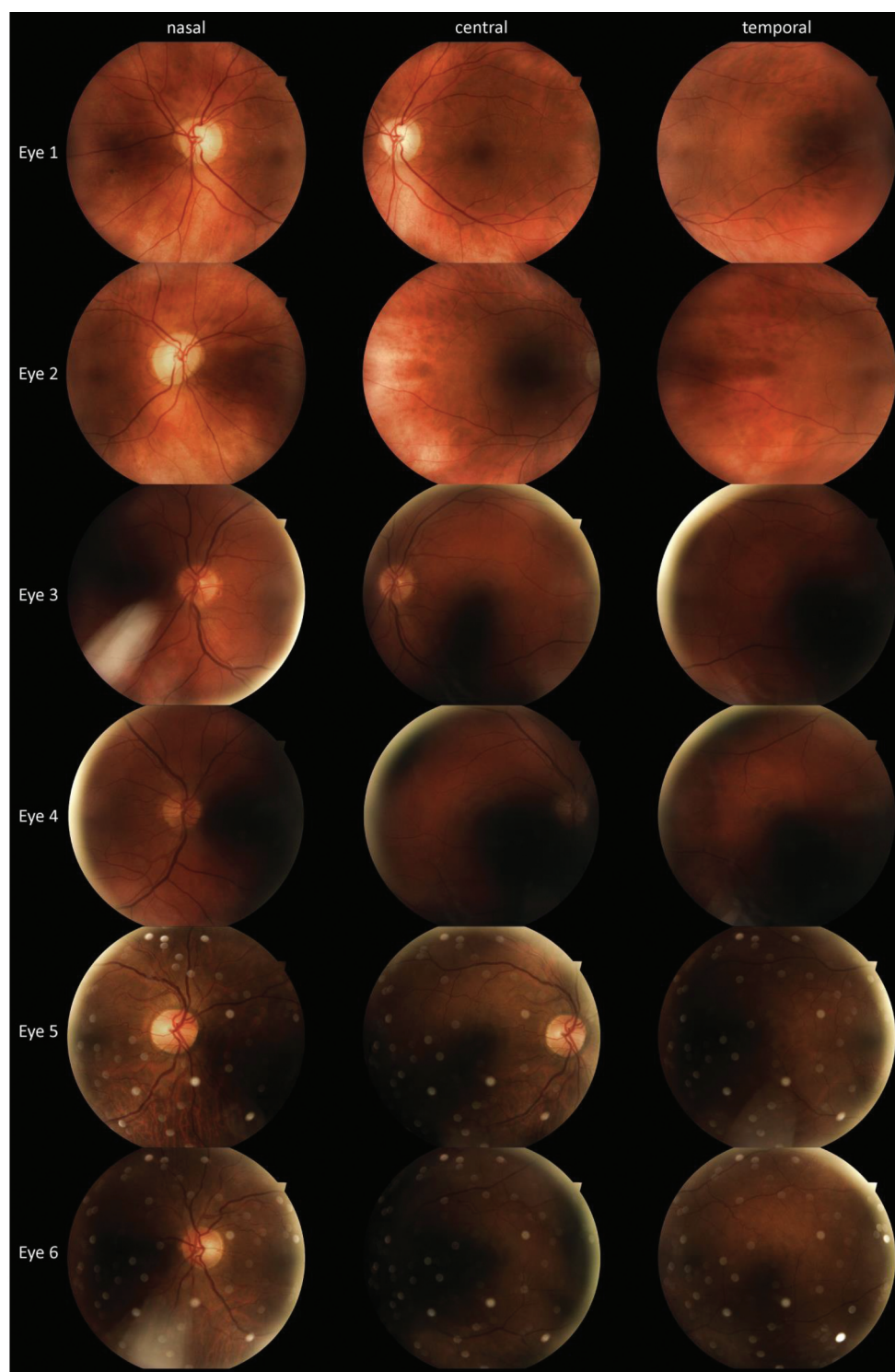


Figure 4. Examples for images with disagreement between human and the automatic algorithm in terms of non-evaluable grading. In a minority of cases, the image quality grading by the algorithm was not understandable from a human perspective: eyes 1 and 2 were graded “non-evaluable” by algorithm but evaluable by the human. Eyes 3–6 were graded “non-evaluable” by the human but “evaluable” by algorithm.

a much larger number of participants for an ophthalmologist check-up compared to manual tele-ophthalmologic assessment of the images. This was based on a larger proportion of images being rated non-

evaluable and reflects the inability of algorithms to deal with real-world sub-optimal image quality as well as images too dissimilar to the algorithm image training set.^{41,42} Our results are similar to other studies

Table 3. Accuracy of the automated image analysis algorithm.

		In the subset of patients determined evaluable by the ophthalmologist (n = 63)	In the subset of patients determined evaluable by the algorithm (n = 48)
Detection of any DED	Sensitivity	0.67 (0.22–0.96)	1.00 (0.40–1.00)
	Specificity	0.74 (0.60–0.84)	0.98 (0.88–1.00)
Detection of referral-warranted DED	Sensitivity	0.50 (0.12–0.88)	0.75 (0.19–0.99)
	Specificity	0.74 (0.60–0.84)	0.98 (0.88–1.00)

DED = diabetic eye disease; values in () represent the 95% confidence interval.

evaluating automated retinal image analysis for DED screening, in which between 14% and 36% of images were rated non-evaluable by the algorithm and led to additional referrals for an ophthalmologist check-up.^{43–45} Our results indicate that utilization of an automated algorithm for DED screening could increase referral rate due to suboptimal image quality and non-evaluable images; however, given our limited sample size, the results should be interpreted with caution. Sensitivity and specificity in our study are similar to studies evaluating clinical or manual grading against several different automated image analysis algorithms in a real-world setting and are largely comparable but not clearly superior to manual grading.^{42–44} Thus, manual grading might still be required for the foreseeable future for challenging images or cases classified as non-evaluable by existing algorithms as well as for quality control.

Frequently observed image artifacts in our study were due to dust particles, smudges (most likely on the camera lens), and eyelashes, which should be preventable by improved training of the examiners (more frequent cleaning of the camera lens and assisting the patient in lifting his/her upper eyelid). Furthermore, illumination and peripheral light artifacts were additional frequent reasons for impaired image quality, and both should be (at least in part) preventable by optimizing the patient's eye alignment with the camera and fixation. Special attention should be given to the elderly and when acquiring images of the region temporal of the macular, as both were associated with impaired image quality. As impaired image quality is a frequent barrier for image analysis using AI, there is a necessity to develop quality standards for retinal imaging.³² The mean test duration time per patient was longer than expected when comparing our results with the literature. This might be explained by discrepancies in the definition of the recorded time interval. The recorded test duration included the time required for the preparation of the patient, repeated image acquisition in order to achieve

the best possible image quality, as well as the breaks needed by the patients during examination.

Strengths of the study include the assessment of feasibility in a real-world setting, obtaining healthcare provider and patient feedback, standardized and comprehensive assessment of image quality, manual double grading of retinal images, and automated image analysis. Limitations include the small number of practices, healthcare providers and patients, especially the small number of patients with DED, the lack of comparators in terms of retinal imaging devices and AI algorithms, and the lack of direct comparison to standard care, i.e., GP referral for an ophthalmologist check-up. The prevalence of DED in our sample was lower than expected.⁴⁶ This might be explained by a selection bias, as patients with known DED may not be as prone to participate in this study as patients without any information on their potential DED. Also, patients with DM primarily managed by their GP and not an endocrinologist might not have an advanced disease, also biasing our results towards a lower rate of DED. It can be beneficial to perform DED classification on an eye and not patient level, as relevant information can still be present in one eye in case image quality is insufficient in the other eye. However, as this DED classification method was used by the automated algorithm, we had to use the same method for the manual classification to allow for comparison with the automated algorithm. Having established the overall feasibility of using the DRS camera in a GP practice with medical assistants obtaining images after a short training, subsequent studies will now assess comparability to standard care including resource use.

In conclusion, our study demonstrated that both GP practice staff and patients prefer a one-stop-solution for DED screening located in the GP practice, that gradable images can be obtained for the majority of patients, and that these are of sufficient quality for automated image analysis. Based on these results, larger implementation studies also including a standard care arm are warranted.

ORCID

Maximilian W.M. Wintergerst  <http://orcid.org/0000-0002-2766-7038>

Robert P. Finger  <http://orcid.org/0000-0003-4253-7597>

Declaration of interests statement

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